

## Paper IV

Apparatus to printed pp. 65–74

Prompt 01. French source edition and independent English translation.

### Witness and editorial treatment

The controlling witness is the attached exact-scope PDF, leaves 3–12, corresponding to full-volume PDF pages 82–91. All ten inherited R27 scan leaves agree with these authority pages at 72 dpi. R27 is a comparator, not the authority; its original bytes remain preserved. Pages 63–64 are copy matter, not author prose.

The French preserves the printed wording, notation, quoted statements, display grouping, and page seams. The English preserves all arguments and mathematical relations, including the legible source problems described below. No conjectural repair is inserted into either text. Running heads and the imprint are translated in the English reader. The author’s note remains a footnote; editorial comments appear only here.

### P. 66: unqualified converse

In the paragraph beginning *Il est facile de voir*, the print states a converse from solvability for every prime factor of  $B$  to solvability for  $B$  itself. The unrestricted statement is retained. It needs qualification when repeated prime factors occur: for example,  $A = 3$ ,  $B = 9$  gives a solution modulo the prime factor 3, but no solution modulo 9, whose fourth-power residues are 0, 1, 4, 7. This counterexample is an editorial calculation, not text supplied by the witness.

Authority: exact-scope leaf 4 / full-PDF page 83, second paragraph.

### P. 70: the printed plus sign

In the paragraph beginning *Le produit*, the print calls  $K$  a divisor of  $\varphi'^2 + 2u'^2$ . The plus sign is legible. Earlier on the same page, the factorization uses  $\varphi'^2 - 2u'^2$ . The plus is reproduced in both editions. The surrounding argument indicates a likely printed error; that diagnosis is editorial and does not change the diplomatic reading.

Authority: exact-scope leaf 8 / full-PDF page 87, the paragraph before the quoted theorem.

## P. 71: restoration of the author’s note

The inherited formula  $r = \rho^2 \mp 1$  is an algebraic rewriting, not the printed expression. The witness reads  $r = b\sigma^2 \pm 1$ , followed by  $s = \rho\sigma$ . The two signs reduced to  $+$  in R27’s norm identity are also printed as  $\pm$ :

$$(rt \pm bsu)^2 - b(ru \pm st)^2 = p.$$

Both editions restore the printed forms. The source marker is  $*$ ), attached after *impair* (*odd* in English), before the sentence’s full stop. R27’s extra closing parenthesis in the French body is not present in the source. The book title *Théorie des Nombres* is italic in the source; it is rendered as *Theory of Numbers* in English. The citation’s printed numbering is retained.

Authority: exact-scope leaf 9 / full-PDF page 88, complete footnote.

Detail witness: [qa/source/p071\\_footnote\\_detail.png](qa/source/p071_footnote_detail.png).

## Other page records

Pages 65, 67–69, and 72–74 require no conjectural emendation. The separate pagewise apparatus files record the title and rules, the lexical reading *de théorèmes* on p.68, lower-case *d’où* on p.69, the equation labels and repeated factor arrays, and the exact continuation points. The catalogue has one record for each of the ten author pages.

The sentence at the foot of p.67 resumes at the head of p.68; that at the foot of p.71 resumes with the result labelled  $(\alpha)$  on p.72. The reciprocity display announced at the end of p.72 begins p.73. The substitution sentence at the foot of p.73 resumes with its power of  $u$  on p.74. No bridging prose is supplied.

Section 4 begins on p.74 and continues beyond this batch. The reader stops after the two-by-two factor display on p.74. This is the Prompt 01 boundary, not the end of Paper IV. Pages 75–98 are not accepted by this checkpoint.

## Layout record

The readers are re-typeset, not photographic facsimiles. Original page boundaries, paragraph starts, display membership, formula punctuation, and the author-note anchor are retained. Physical within-page line wrapping and justification are not copied; line-end word division is reflowed, without changing lexical hyphens such as *tout-à-fait* and *à-peu-près*. These layout choices are explicit, not silent lexical normalization. The untouched authority and page images remain the reference for the original lineation and letterforms.